

Null-Space Interaction: Enabling Reaching Task During Typing via Shoulder Movements

Mingyo Kim*
The University of Tokyo

Takashi Ota†
The University of Tokyo

Takuji Narumi‡
The University of Tokyo

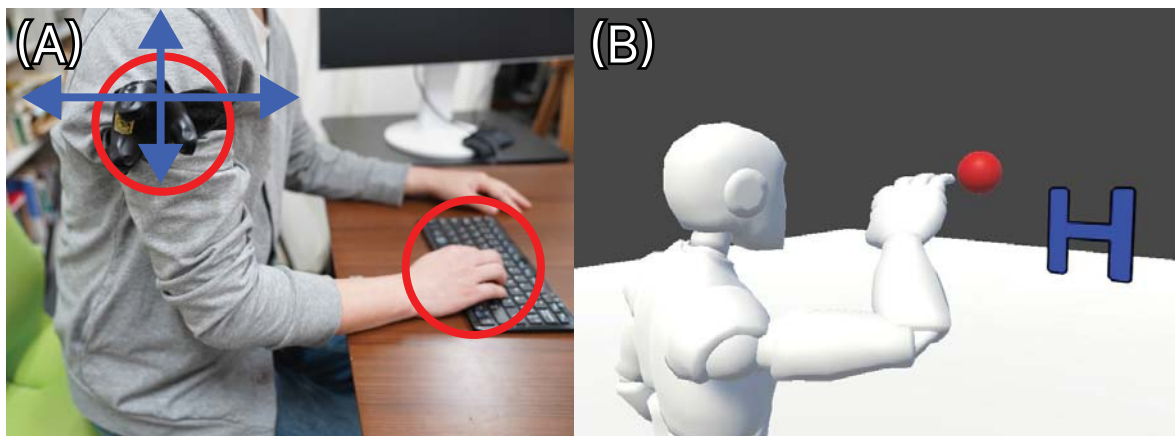


Figure 1: (A) While typing on the keyboard, user can move the shoulder in the sagittal plane originating from the shoulder to manipulate the arm. (B) By manipulating the arm, user has to reach the target (red ball) while typing the letter on the screen.

ABSTRACT

In workspaces, users often engage in manual tasks such as typing, where their hands are physically anchored to a device. Existing interaction paradigms typically require those users to lift their hands for mid-air gestures to interact with distant virtual objects, causing context switching costs and physical fatigue. However, the human upper limb possesses kinematic redundancy, meaning the proximal joint can move independently within the kinematic null space even while the end-effector remains fixed. Leveraging this biomechanical characteristic, we propose an interaction interface that utilizes shoulder movements to control a virtual arm for reaching tasks, while keeping the hand on the primary task.

Our system maps the subtle protraction and elevation of the shoulder to the virtual hand's position using a linear amplification gain, allowing users to reach objects without disengaging from their primary manual task. To validate the feasibility of this approach, we designed a dual-task user study comparing unimanual typing performance (primary task) with and without the concurrent shoulder-based reaching task (secondary task). We analyze key metrics including typing speed, reaching time, and trajectory of virtual hand. This work explores the potential of utilizing the body's unused degrees of freedom to expand interaction capabilities, aiming to achieve seamless parallel processing of manual and spatial tasks.

Index Terms: Shoulder Interface, Multitasking, Kinematic Redundancy, Extra Limbs

*e-mail: mingyo-kim@cyber.t.u-tokyo.ac.jp

†e-mail: ota@cyber.t.u-tokyo.ac.jp

‡e-mail: narumi@cyber.t.u-tokyo.ac.jp

1 INTRODUCTION

For many daily tasks where the hand is anchored, such as typing on a keyboard or resting on a surface, the proximal joints retain a range of motion within the kinematic null space that remains largely unutilized. This biomechanical redundancy highlights a fundamental inefficiency in current interface paradigms, which typically enforce a restrictive “one-limb, one-task” rule. By harvesting this latent motor capacity, we can decouple the functional workspace from the physical body, creating an interface paradigm applicable to both extending reach in Virtual Reality and controlling supernumerary robotic limbs in the physical world.

Moreover, relying solely on the distal end-effector for every interaction is biomechanically expensive. For simple tasks that do not require high degrees of freedom—such as selecting a target or triggering a binary event—mobilizing the entire kinematic chain is disproportionate to the task's complexity. This unnecessary recruitment of the arm forces users to sustain anti-gravity postures, directly contributing to the fatigue and limiting the duration of comfortable interaction.

In this paper, we propose an interaction technique that repurposes these surplus DoFs as an auxiliary input channel for multitasking. By decoupling the virtual arm's reach from the physical hand's position, our approach utilizes subtle shoulder movements to control distal interactions. This enables users to perform reaching tasks within the virtual workspace using shoulder protraction or elevation, all while their physical hands remain engaged in a primary task.

This decoupling strategy not only facilitates parallel processing but also offers significant ergonomic and accessibility benefits. By substituting large, anti-gravity arm movements with minimal proximal actuation, the system inherently mitigates the fatigue. Furthermore, it lowers the physical barrier for interaction, offering an adaptable interface for users with limited range of motion.

Our primary contribution is the initial assessment of this concept through a preliminary user study, which demonstrates the feasibility and effectiveness of leveraging shoulder from redundant DoFs to

support simultaneous dual-task performance.

2 RELATED WORKS

2.1 Motion gain and micro gestures

High Control-Display (C-D) gain is a critical mechanism for minimizing physical fatigue in VR by enabling extensive virtual manipulations through subtle physical inputs, or micro-movements [4]. While high gain can potentially amplify motor noise, the human visuomotor system demonstrates remarkable plasticity, rapidly recalibrating to gain perturbations [5] and maintaining performance even under dynamic gain variations [3]. This adaptability supports the feasibility of using macroscopic, proximal body parts for precise control.

Research has challenged the dominance of distal joints (e.g., fingers) for precise input. Balakrishnan and MacKenzie revealed that proximal segments like the forearm possess information processing bandwidths comparable to or exceeding those of the fingers [1]. Also, recent interfaces have successfully utilized non-manual body parts—such as subtle head gestures [7] or torso leaning [11]—to augment and support interaction capabilities.

Building on this body of work, we propose the shoulder as an independent input channel, allowing users to perform decoupled reaching tasks efficiently without disrupting primary manual activities.

2.2 Redundant DoF as an Interface

The human kinematic chain is inherently redundant, possessing more degrees of freedom (DoF) than are strictly required to specify the position and orientation of an end-effector. Bernstein postulated that this surplus is not a computational burden, but a critical resource for motor equivalence, allowing the central nervous system to achieve a consistent task goal through infinite variations of joint configurations [2]. In robotics, this redundancy is mathematically described as the null space, which allows a manipulator to perform secondary tasks—such as obstacle avoidance or torque optimization—without altering the primary end-effector's pose [9]. Similarly, in biological control, this null space provides a buffer for minimizing fatigue and optimizing biomechanical comfort without disrupting the main task.

This principle has been successfully applied in Body-Machine Interfaces for assistive technologies. For instance, Jain et al. demonstrated that individuals with severe motor impairments could effectively control high-DoF robotic arms by remapping residual motions from proximal joints [8].

These findings suggest that the brain can rapidly reorganize its control strategy by projecting complex body movements onto a low-dimensional control subspace, effectively transforming physical redundancy into functional synergies.

Shimobayashi et al. also explored the potential of exploiting upper limb redundancy, demonstrating that the shoulder can serve as an independent input channel to control supernumerary appendages while the hands are occupied [10]. However, their study primarily focused on static holding tasks, where the hands remain fixed in a grasping posture. In contrast, our work addresses the more challenging context of dynamic manual tasks, specifically continuous typing, which introduces constant kinematic noise and demands higher stability. Furthermore, while their approach relies on the discrete classification, our system utilizes continuous tracking of shoulder movements to enable fluid and precise analog control of the virtual arm.

3 SYSTEM IMPLEMENTATION

The system presented in this study was developed using the Unity engine (version 2022.3 LTS) with a VIVE Tracker 3.0.

3.1 Calibration

Since the user's physical hands are engaged in typing and are not tracked visually in the VR environment, precise control relies on a calibration process to account for individual differences in typing posture and Range of Motion (ROM). The calibration sequence consists of two stages designed to establish a reference origin and physical boundaries:

- **Relax** : The user maintains a comfortable posture for the primary task. The system records the shoulder tracker's position ($\mathbf{P}_{relax}$) and rotation ($\mathbf{R}_{relax}$) as the reference origin.
- **ROM Measurement** : The user is guided to move their shoulder to its maximum extents in four directions (Forward, Backward, Up, and Down). The system measures the maximum displacement from the origin in each direction relative to the relaxed rotation, storing these values as physical boundaries (x_{min}, x_{max}) for each axis.

3.2 Manipulation and Control

To enable precise control of the virtual arm using shoulder micro-movements, we implemented an interface that maps the shoulder's position to the target of an Inverse Kinematics (IK) solver. During runtime, the system calculates the shoulder's current relative displacement ($\Delta \mathbf{P}_{curr}$) from the relaxed position. To ensure intuitive control aligned with the user's body orientation, the raw world position is transformed into the local coordinate system of the relaxed pose by Equation 1.

$$\Delta \mathbf{P}_{local} = \mathbf{R}_{relax}^{-1} \cdot (\mathbf{P}_{curr} - \mathbf{P}_{relax}) \quad (1)$$

Considering the biomechanical constraints and the task requirements, we limited the control of the arm to two axes: Anterior-Posterior (Z-axis) and Superior-Inferior (Y-axis). Lateral movements were excluded due to the practical DoF of the shoulder during typing. To map the small physical range of the shoulder to the larger workspace of the virtual hand, we employ a linear mapping function. First, the local physical displacement x is normalized to a coefficient $\alpha \in [0, 1]$ using Equation 2 based on the calibrated physical boundaries (x_{min}, x_{max}).

$$\alpha = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (2)$$

where x represents the current displacement ($\Delta \mathbf{P}_{local}$) along a specific axis. This normalized value is then mapped to the target virtual workspace range ($[y_{min}, y_{max}]$). Finally, the target position ($\mathbf{P}_{target}$) is determined by Equation 3, which drives the IK solver to position the virtual hand.

$$\mathbf{P}_{target} = \mathbf{y}_{min} + \alpha \cdot (\mathbf{y}_{max} - \mathbf{y}_{min}) \quad (3)$$

This linear amplification allows movements of the shoulder to effectively control the reach of the virtual hand on the anterior vertical plane originating from the shoulder while keeping the physical hands stationary.

4 USER STUDY

To evaluate the feasibility and efficiency of the proposed shoulder-based interface, we conducted a within-subject user study employing a dual-task paradigm. The experiment participants consisted of three male individuals aged 22 to 26, who were invited to participate after providing their consent. All participants said they are familiar with the qwerty layout of the keyboard.

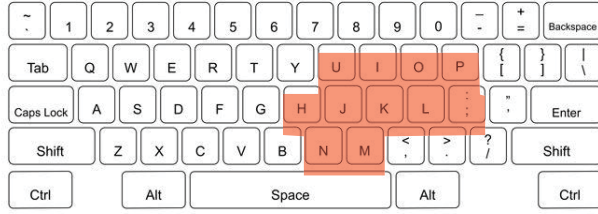


Figure 2: Random input was selected among the red area of the qwerty layout keyboard for the unimanual typing task.

4.1 Design

The primary objective was to verify whether users could effectively control a virtual arm using shoulder micro-movements for the secondary task without significantly impairing their performance in a continuous primary task. The experiment simulates a scenario where the user's right hand is fully occupied. For the primary task, participants use their right hand to continuously type random characters generated from a subset of the QWERTY layout shown in Figure 2. Immediate color-coded visual feedback is provided upon each keystroke to maintain a consistent cognitive load. Since users cannot visually monitor their hands, this feedback serves as a safeguard, offering an immediate opportunity for correction and preventing a single slip from escalating into a stream of subsequent typos.

Concurrently, the secondary task requires participants to intercept spherical targets that appear at random intervals (4–8 seconds) within the reachable workspace (radius: 0.8m, $\pm 30^\circ$). Also this reachable target ensures the target key remains within the field of view during the reaching task. Participants are instructed to touch these targets using the virtual hand—controlled via shoulder micro-movements—while strictly keeping their physical right hand anchored to the keyboard.

The study is divided into two 60-second phases: a Baseline Phase (typing only) to establish standard performance, and a Dual-Task Phase (simultaneous typing and reaching) to observe adaptation and interference effects. After enough practice with the Baseline Phase to adapt to the keyboard, user performed a single Baseline Phase and three Dual-Task Phases.

4.2 Data Acquisition and Metrics

A logging system records quantitative data to assess both task performance and physical behavior. Typing performance is evaluated based on Typing Speed (KPM, Keystroke Per Minute) and Error Rate, comparing the Baseline and Dual-Task phases to quantify any interference caused by the shoulder interaction. Reaching performance is measured by Reaching Time and its efficiency. Crucially, to validate the capability of our interface, the system tracks the Physical Hand Drift, defined as the spatial displacement of the user's physical right hand during the reaching action. Minimal drift would quantitatively confirm that the user can control the virtual arm independently without disrupting the posture required for the manual task.

4.3 Result

4.3.1 Accuracy and Speed of the typing

We observed the participants' typing speed (Keystroke Per Minute, KPM) and accuracy across the Baseline and three Dual-Task trials. As shown in Figure 3, the average typing speed showed an initial decrease in Trial 1 (67.20 KPM) compared to the Baseline (76.52 KPM). Although performance dipped in Trial 1, the upward

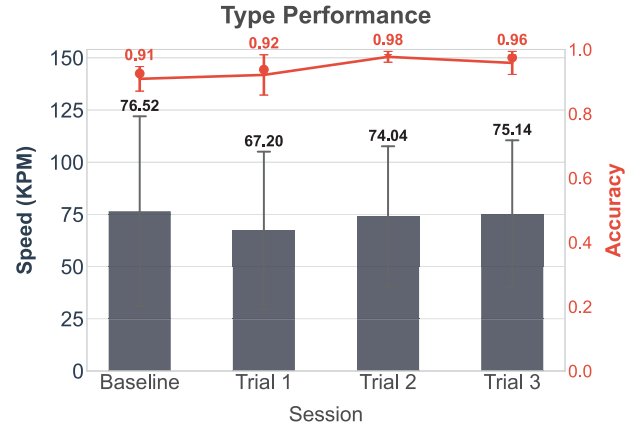


Figure 3: Average typing speed (Keystroke Per Minute, KPM) and accuracy across the Baseline and three Dual-Task trials. The bar chart represents speed, and the line plot represents accuracy.

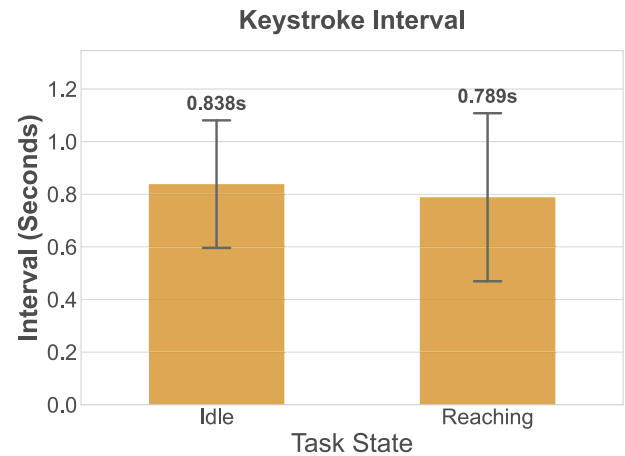


Figure 4: Comparison of average keystroke intervals during the dual-task condition. "Idle" indicates keystroke intervals during reach target is hidden and "Reaching" (dual-task) states keystroke intervals while reach target is shown.

trend toward baseline observed in Trials 2 and 3 suggests the potential that participants were likely overcoming the initial interference through an adaptation process.

Notably, typing accuracy showed an increasing trend throughout the sessions. This counter-intuitive improvement might be explained by the Yerkes-Dodson Law [14], which posits that performance improves with physiological or mental arousal up to an optimal point. The addition of the secondary reaching task may have elevated the participants' arousal from a potentially monotonous state, thereby enhancing their focus on the primary typing task.

Regarding the cognitive Cost, Figure 4 illustrates the difference in average keystroke intervals between "Reaching" states (0.789s) and "Idle" states (0.838s) shows no significant difference for this user study. The relatively small delay suggests that the shoulder-based interaction might be performed in parallel with typing without causing a significant disruption to the motor rhythm.

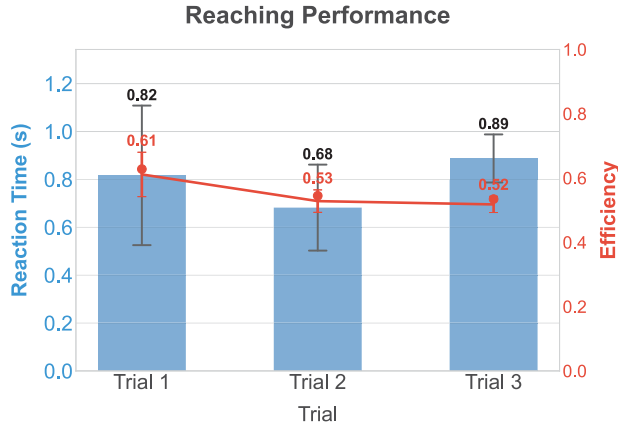


Figure 5: Performance metrics for the secondary reaching task across trials. Blue bars indicate average Reaction Time for the reach target (s), and the red line indicates Path Efficiency, the ratio of the ideal distance to the actual traveled path length.

4.3.2 Response time and Efficiency of the reaching

For the secondary reaching task, we analyzed Reaction Time and Path Efficiency across trials. Path efficiency is defined as the ratio of the Euclidean distance between the hand's position at target onset and the target, to the actual distance traveled. As observed on Figure 5, the average Reaction Time fluctuated across sessions. Rather than indicating a lack of learning, these fluctuations likely reflect the varying difficulty levels associated with the randomized target positions generated in each trial. Since some targets may require more complex shoulder maneuvering than others, the average time is naturally susceptible to the specific spatial distribution of targets in a given session. However, the observed decrease in standard deviation across trials points to the possibility of user performance becoming more stable.

Regarding path efficiency, the values remained in the range of 0.5 to 0.6 which indicate curved trajectories. As highlighted by Yang et al., redundancy allows equivalent joint combinations to minimize path deviations [13]. By using a shoulder to control, the system lacks the compensatory mechanisms provided by multi-joint synergy, which likely contributes to the increased curvature and reduced path efficiency.

However, it is difficult to definitively categorize this as suboptimal performance. It has been observed that human hand trajectories during reaching do not follow a strictly straight-line path. The research conducted by Xiao et al. quantifies this non-linearity through the Reach Path Ratio (RPR), showing that the actual path length exceeds the shortest straight-line distance[12]. More specifically, the degree of curvature in these motions can vary depending on the spatial location of the target[6]. Determining the true efficiency of our interface requires further comparison with physical hand reaching data under the same condition in future studies.

4.4 User Feedback

To complement the quantitative performance metrics, we collected qualitative feedback from the participants regarding their subjective experiences during the dual-task sessions. The most consistent observation was the distinct separation between visual attention and physical motor control. Participants noted that accurately perceiving the 3D position of the target sphere required an explicit shift of visual gaze away from the typing interface. This momentary saccade introduced a cognitive interruption, making the experience feel less like seamless parallel processing and more like a rapid,

high-frequency task switching scenario. However, despite this visual demand, participants emphasized that the physical overhead associated with this transition was negligible. Since their hands remained physically anchored to the keyboard throughout the reaching task, they could resume typing immediately after the visual check without needing to re-orient their hand position or posture.

Furthermore, the shoulder-based interface itself was reported to be physically unobtrusive. Participants reported they did not feel that the shoulder micro-movements interfered with the fine motor control required for keystrokes, nor did they experience significant muscle fatigue or discomfort during the trials.

This qualitative feedback suggests that while visual attention management remains a bottleneck, our system might achieve its primary goal of kinematic decoupling, allowing for efficient multi-tasking with minimal physical intrusion.

5 CONCLUSION

In this study, we proposed a shoulder-based virtual arm control interface designed to enable distal reaching tasks with minimal disruption to manual workflows. Drawing on the concept of kinematic null space, our approach exploits the shoulder's role as a redundant degree of freedom during manual tasks like typing. By leveraging this biomechanical characteristic, the system allows the proximal joint to control a virtual arm independently, effectively decoupling the reaching motion from the stationary hand.

We conducted a preliminary dual-task user study with a small group of three participants to evaluate the feasibility of this approach. The results suggested the potential that participants could successfully manipulate the virtual hand to reach distant targets, despite the significant discrepancy in the range of motion between the physical shoulder and the virtual arm. Furthermore, our analysis indicated that this mapping strategy has potential to effectively minimized unintentional displacement of the physical hands, ensuring physical stability and allowing users to maintain their typing rhythm with minimal interference.

Moving forward, we plan to enhance the ecological validity of our research by introducing more complex and practical tasks, such as drag-and-drop operations or GUI manipulations, to verify the system's utility in realistic work scenarios. Furthermore, we aim to investigate the residual degrees of freedom in other body parts beyond the shoulder. By identifying and utilizing these available DoFs, we hope to expand this paradigm into seamless, low-cost interaction environment in VR.

ACKNOWLEDGMENTS

This work was partially supported by JSPS KAKENHI (JP25K03162), JST Moonshot Research & Development Program (JPMJMS2013), and JST ASPIRE (JPMJAP2404).

REFERENCES

- [1] R. Balakrishnan and I. S. MacKenzie. Performance differences in the fingers, wrist, and forearm in computer input control. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*, CHI '97, p. 303–310. Association for Computing Machinery, New York, NY, USA, 1997. doi: 10.1145/258549.258764 2
- [2] N. Bernstein. *The Co-ordination and Regulation of Movements*. Pergamon Press, 1967. 2
- [3] R. Blanch, Y. Guiard, and M. Beaudouin-Lafon. Semantic pointing: Improving target acquisition with control-display ratio adaptation. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '04, pp. 519–526. Association for Computing Machinery, New York, NY, USA, Apr. 2004. doi: 10.1145/985692.985758 2
- [4] G. Casiez, D. Vogel, R. Balakrishnan, and A. Cockburn. The Impact of Control-Display Gain on User Performance in Pointing Tasks. *Human-Computer Interaction*, 23(3):215–250, Aug. 2008. doi: 10.1080/07370020802278163 2

- [5] E. J. Cohen, K. Wei, and D. Minciacci. Visuomotor perturbation in a continuous circle tracing task: Novel approach for quantifying motor adaptation. *Scientific Reports*, 9(1):18679, Dec. 2019. doi: 10.1038/s41598-019-55241-4 2
- [6] J. Faraway. Modeling hand trajectories during reaching motions. *submitted for publication to Journal of the American Statistical Association*, 2001. 4
- [7] B. J. Hou, J. Newn, L. Sidenmark, A. A. Khan, and H. Gellersen. Gazeswitch: Automatic eye-head mode switching for optimised hands-free pointing. *Proc. ACM Hum.-Comput. Interact.*, 8(ETRA), May 2024. doi: 10.1145/3655601 2
- [8] S. Jain, A. Farshchiansadegh, A. Broad, F. Abdollahi, F. Mussa-Ivaldi, and B. Argall. Assistive robotic manipulation through shared autonomy and a body-machine interface. In *2015 IEEE International Conference on Rehabilitation Robotics (ICORR)*, pp. 526–531, 2015. doi: 10.1109/ICORR.2015.7281253 2
- [9] Y. Nakamura, H. Hanafusa, and T. Yoshikawa. Task-Priority Based Redundancy Control of Robot Manipulators. *The International Journal of Robotics Research*, 6(2):3–15, June 1987. doi: 10.1177/027836498700600201 2
- [10] H. Shimobayashi, T. Sasaki, A. Horie, R. Arakawa, Z. Kashino, and M. Inami. Independent Control of Supernumerary Appendages Exploiting Upper Limb Redundancy. In *Proceedings of the Augmented Humans International Conference 2021*, AHs '21, pp. 19–30. Association for Computing Machinery, New York, NY, USA, July 2021. doi: 10.1145/3458709.3458980 2
- [11] O. Špakov, P. Isokoski, and P. Majaranta. Look and lean: accurate head-assisted eye pointing. In *Proceedings of the Symposium on Eye Tracking Research and Applications*, ETRA '14, p. 35–42. Association for Computing Machinery, New York, NY, USA, 2014. doi: 10.1145/2578153.2578157 2
- [12] X. Xiao, H. Hu, L. Li, and L. Li. Comparison of dominant hand to non-dominant hand in conduction of reaching task from 3d kinematic data: Trade-off between successful rate and movement efficiency. *Mathematical biosciences and engineering*, 16(3):1611–1624, 2019. 4
- [13] J.-F. Yang, J. P. Scholz, and M. L. Latash. The role of kinematic redundancy in adaptation of reaching. *Experimental brain research*, 176(1):54–69, 2007. 4
- [14] R. M. Yerkes and J. D. Dodson. The relation of strength of stimulus to rapidity of habit-formation. *Journal of Comparative Neurology and Psychology*, 18(5):459–482, 1908. doi: 10.1002/cne.920180503 3